

YONGHOON JEONG

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CONTACT

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RESEARCH INTERESTS

- **Ultra-Low Power Wide-Area Network (LPWAN)**
- **Ocean IoT**
- **Communication System Design**
- (Previous) Autonomous Vehicles – 3D Perception with Deep Learning
- (Previous) Biomedical Signal Processing – Remote Photoplethysmography (rPPG) with Deep Learning
- (Previous) Operating Systems in Software-Defined Vehicles (SDV)
- (Previous) Direct Memory Access (DMA) Virtualization

EDUCATION

- | | |
|--|-----------------------|
| M.S. in Electrical and Computer Engineering | Mar. 2024 – Feb. 2026 |
| - Seoul National University, Seoul, Republic of Korea | |
| - Advisor: Prof. Sae-woong Bahk | |
| B.S. in Electronics and Communications Engineering | Mar. 2018 - Aug. 2023 |
| - Kwangwoon University, Seoul, Republic of Korea | |
| - Advisor: Emeritus Prof. Young-chul Chung and Prof. Young-seok Choi | |

CAREERS

- | | |
|---|-----------------------|
| Electronics and Telecommunications Research Institute (ETRI), Research Intern | Jan. 2023 – Feb. 2023 |
| - Developed the virtualization model for DMAC Integrated AI Hardware | |
| - Smart Manufacturing Application Laboratory, Seongnam, Republic of Korea | |
| - Advisor: Won-jong Kim | |
| Korea Advanced Institute of Science and Technology (KAIST), Research Intern | Jun. 2022 – Aug. 2022 |
| - Perception for autonomous high-speed driving to address ‘false positive’ issue, while preparing for the Indy Autonomous Challenge (IAC) at CES 2023 | |
| - Unmanned System Research Group (USRG) Laboratory, Daejeon, Republic of Korea | |
| - Advisor: Prof. David Hyun-chul Shim | |
| Software Maestro (SWM, 12th) | Apr. 2021 – Dec. 2021 |
| - Developed an algorithm for object tracking and hazard detection by fusing LiDAR and Camera | |
| - Organized by the Ministry of Science and ICT (MSIT) | |
| Undergraduate Researcher | Sep. 2020 – Aug. 2023 |
| - Biomedical Signal Processing – Remote Photoplethysmography (rPPG) with Deep Learning | |
| - Developed a deep learning model for non-contact heart rate estimation from facial videos. | |
| - Advisor: Prof. Young-seok Choi | |

PUBLICATIONS

International Journals

- “Hyperbolic Frequency Shift Chirp Modulation for Underwater Acoustic Communications”, in preparation.
- “On the Interference Resilience of Non-Linear Chirp Modulation for LoRa: An Analytical Insight into Energy Scattering Effect”, submitted 2026.
- Dongrak Choi, Yubin Choi, **Yonghoon Jeong**, Jeongyeup Paek, and Saewoong Bahk, "D²-PLC: Holistic Design and Implementation of High-Datarate Duplex DC Power Line Communication Network", to appear in *IEEE Internet of Things Journal*, 2025.
- **Yonghoon Jeong**, Youngseok Choi, "Diffusion-Phys: noise-robust heart rate estimation from facial videos via diffusion models", *Biomedical Engineering Letters*, vol.15, pages 575–585, Apr, 2025.

International Conferences

- Dongrak Choi, **Yonghoon Jeong**, Yubin Choi, and Saewoong Bahk, "Demo: A Programmable High-Throughput Duplex DC-PLC Testbed for Power and Data Integration", *IEEE ICNP 2025 demo*, Seoul, Korea, Oct. 19-22, 2025.

Domestic Conferences

- **Yonghoon Jeong**, Saewoong Bahk, “A Study on Direct Current Power Line Communication (DC-PLC) with Applications in Automotive, Maritime, and Aviation,” *The Fall Conference of the Korea Institute of Communications and Information Sciences (KICS)*, Gyeongju, Korea, November 20-22, 2024.
- **Yonghoon Jeong**, “Development of Non-contact Heart Rate Measurement Technology Based on Facial Video,” *The Conference of the Korean Institute of Information Scientists and Engineers (KIISE)*, Jeju, Korea, June 23-25, 2021.

AWARDS / SCHOLARSHIP

Scholarship Recipient	2022–2023
- Korea Scholarship Foundation for the Future Leaders (KOSFFL 52th)	
Excellence Prize	Jul. 2021
- Undergraduate Division, 2021 Korea Computer Conference (KCC 2021)	
Top Prize (University President's Award)	Nov. 2020
- Regional-Focused Capstone Design, organized by Nowon, Seoul	
Scholarship Recipient	2020–2020
- Samsung Scholarship Foundation	
Dean’s List	Apr. 2020
- Highest GPA in Fall 2019 Semester	

PROGRAMMING SKILLS

Programming Languages: C, C++, Python, MATLAB
Frameworks & Libraries: ROS, ROS2, TensorFlow, PyTorch
Simulation Tools: ns-3, MATLAB/Simulink
Development Tools: Linux, Git, Docker

TEACHING EXPERIENCE

Creative Engineering Design , Prof. Sae-woong Bahk	Sep. 2024 – Dec. 2024
Electronic Circuits Experiment 1 , Emeritus Prof. Young-chul Chung	Mar. 2023 – Jun. 2023
Electronic Circuits Experiment 2 , Emeritus Prof. Young-chul Chung	Sep. 2023 – Dec. 2023
Electronic Circuits Experiment 1 , Prof. Young-seok Choi	Mar. 2022 – Jun. 2022